



STI Hackathon 2025: Complete Setup Guide

Welcome to the comprehensive setup guide for STI Hackathon 2025. This step-by-step tutorial will walk you through creating your account, setting up your virtual machine, and deploying applications using Coolify. By the end of this guide, you'll have everything needed to build and deploy your hackathon project successfully.

What You'll Accomplish

01

Create STI Console Account

Register and verify your account on the STI platform console

02

Join Hackathon Tenant

Request access and join the STI Industry 4.0+ tenant

03

Set Up VPN Connection

Configure WireGuard VPN for secure access to resources

04

Deploy Virtual Machine

Create and configure your Ubuntu development environment

05

Connect to Coolify

Set up application deployment platform for easy project hosting

sti.go.ug

Login



[forgot password](#)

LOGIN

[Not Registered? register now](#)

Step 1: Access the STI Console

Navigate to <https://console.sti.go.ug> in your web browser. This is the main portal where you'll manage your hackathon resources and virtual machines.

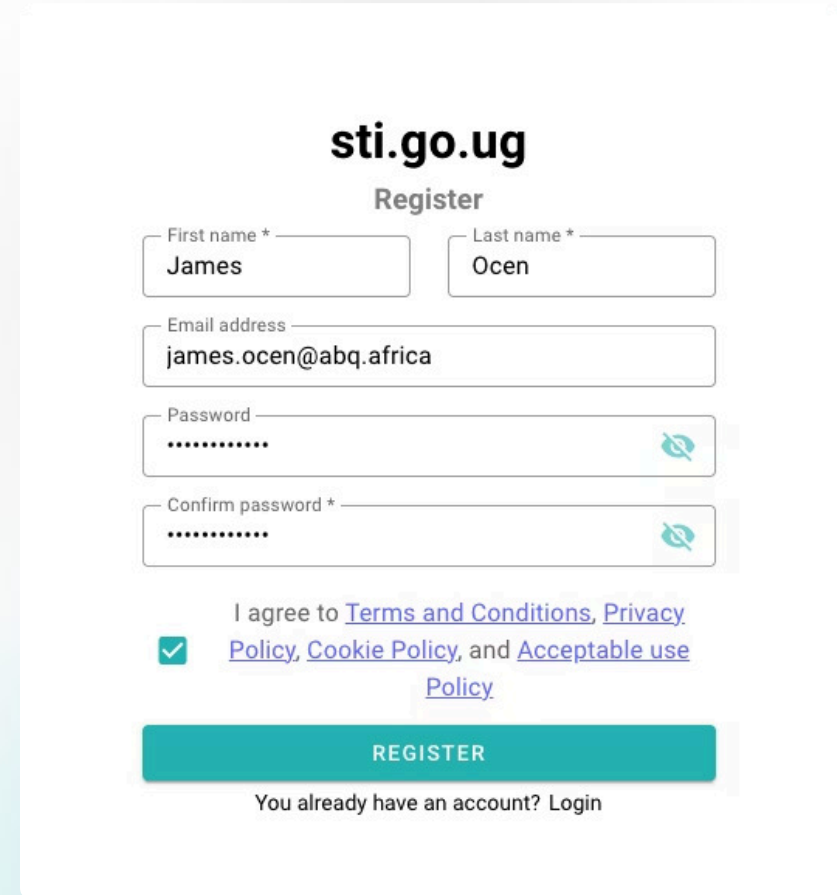
The STI Console provides a comprehensive dashboard for accessing cloud resources, managing virtual machines, and connecting to various services needed for your hackathon project development.

Step 2: Register Your Account

Registration Process

1. Click "Register Now" on the homepage
2. Fill in your personal details accurately
3. Read and accept the terms and conditions
4. Click the "Register" button to submit

📄 **Important:** Use a valid email address as you'll receive verification codes and important notifications about your tenant access throughout the hackathon.



The screenshot shows the registration page for sti.go.ug. At the top, the logo 'sti.go.ug' is displayed in bold black text, followed by the word 'Register' in a smaller, grey font. Below this, there are four input fields: 'First name *' with the value 'James', 'Last name *' with the value 'Ocen', 'Email address' with the value 'james.ocen@abq.africa', and 'Password' with masked characters '.....'. To the right of the password field is a small teal icon of a person with a speech bubble. Below the password field is a 'Confirm password *' field, also with masked characters '.....' and a similar teal icon. Under these fields, there is a checkbox with a teal checkmark, followed by the text 'I agree to' and three links: 'Terms and Conditions', 'Privacy Policy', and 'Acceptable use Policy'. At the bottom, there is a large teal button with the word 'REGISTER' in white capital letters. Below the button, the text 'You already have an account? Login' is displayed in a smaller, grey font.

sti.go.ug

Register

First name * James

Last name * Ocen

Email address james.ocen@abq.africa

Password

Confirm password *

☒ I agree to [Terms and Conditions](#), [Privacy Policy](#), [Cookie Policy](#), and [Acceptable use Policy](#)

REGISTER

You already have an account? Login

Step 3: Email Verification

After registering, check your email for an OTP (One-Time Password) verification code. This security step ensures your account is properly authenticated and prevents unauthorised access.

Enter the OTP code when prompted to complete your registration. Once verified, you'll be automatically redirected to the web console dashboard where you can begin accessing STI services.

i Verification code is sent to your email to validate your ownership. Fill in the code in the box below your email **×**

sti.go.ug

Register

Email address

james.ocen@abq.africa

Code:

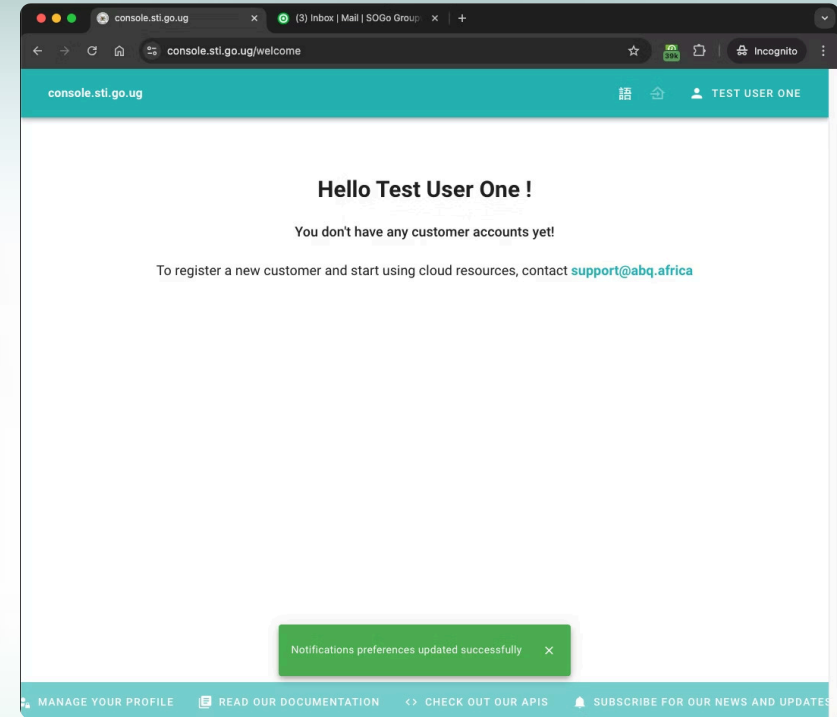
Resend code

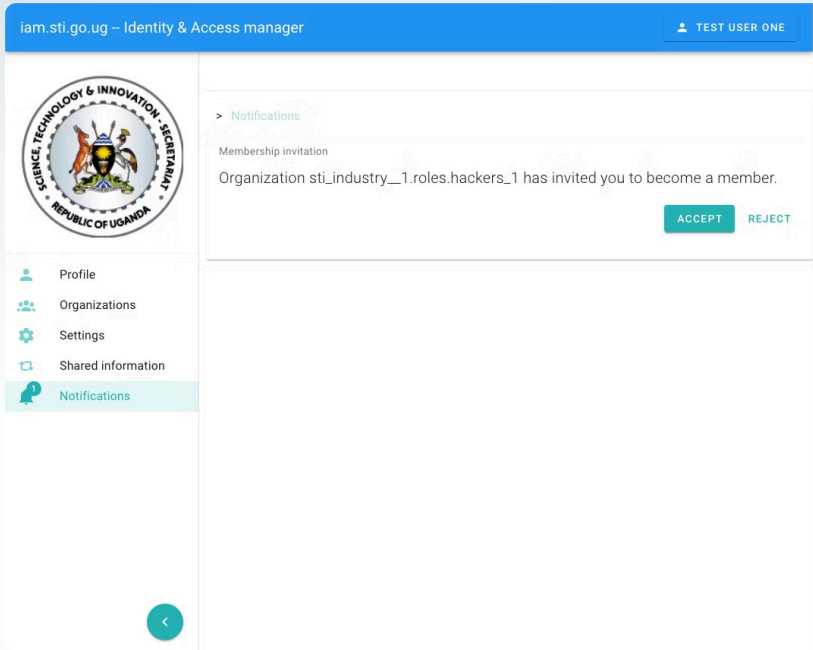
Step 4: Request Tenant Access

Send Support Email

Email support@abq.africa to request invitation to the STI Industry 4.0+ tenant. Include your registered email address and mention you're participating in Hackathon 2025.

The support team will process your request within 24 hours. You'll receive an email notification once you've been successfully added to the hackathon tenant with access to all necessary resources.

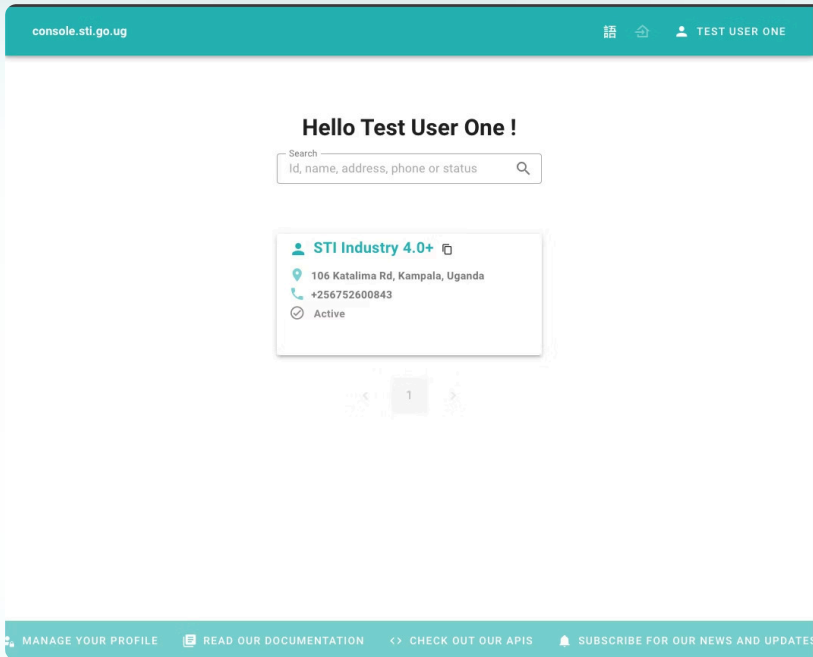




Step 5: Accept Tenant Invitation

When you receive the tenant invitation email, click the provided link to accept access to the STI Industry 4.0+ tenant. This grants you permission to use hackathon resources including virtual machines and development tools.

After accepting, return to the web console and refresh the page. You should now see the "STI Industry 4.0+" tenant available in your dashboard, indicating successful access to hackathon resources.



Step 6: Access Your Tenant

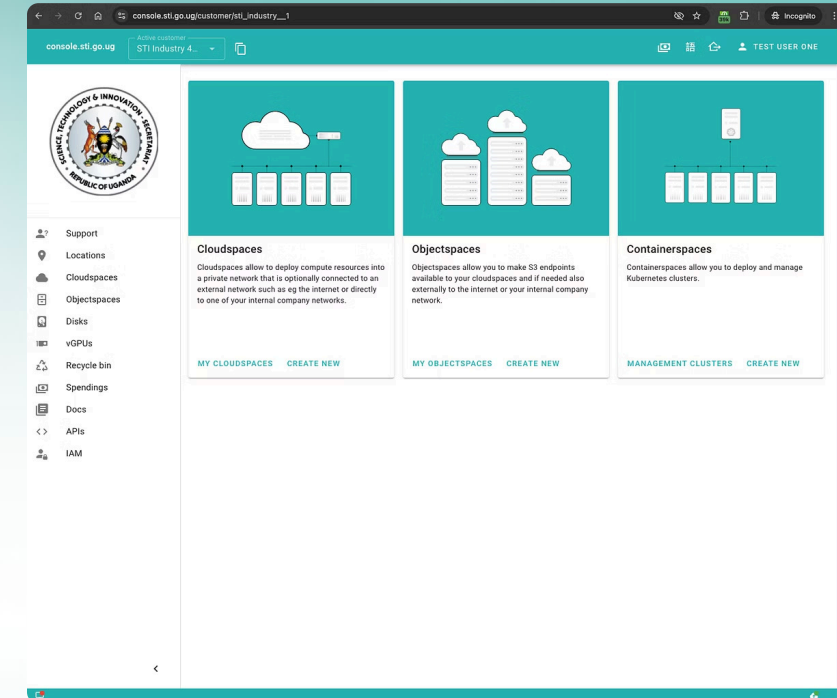
Click on the "STI Industry 4.0+" tenant tile in your dashboard to access its resources and features. This opens up the hackathon-specific environment where you can create virtual machines, manage storage, and deploy applications.

The tenant dashboard provides access to cloudspaces, networking tools, and various services specifically configured for hackathon participants.

Step 7: Navigate to Hackathon Cloudspace

From the tenant dashboard, navigate to **Cloudspaces > hackathon 2025**. This is your dedicated workspace containing all the resources, templates, and configurations specifically prepared for the hackathon event.

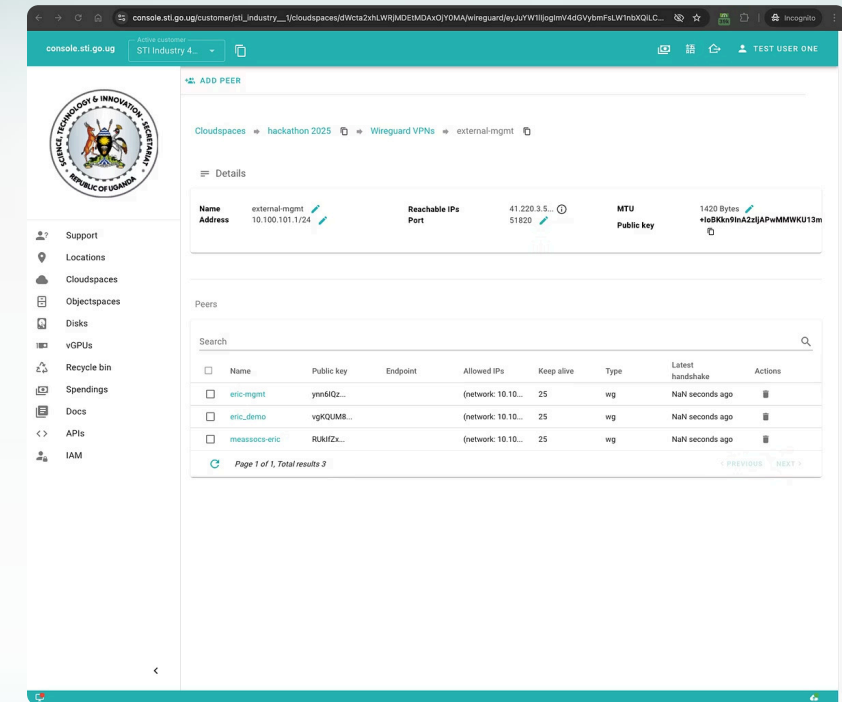
The hackathon cloudspace includes pre-configured networks, security groups, and access to specialised services that will support your project development throughout the event.

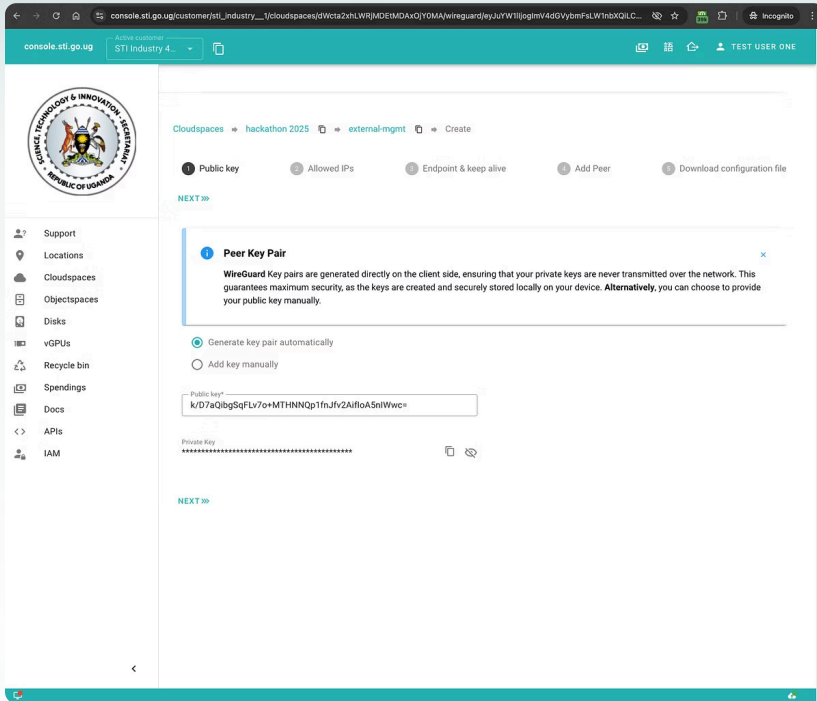


Step 8: Set Up WireGuard VPN

Before creating your virtual machine, you need to establish a secure VPN connection. Navigate to **WireGuard VPNs > external-mgmt** to begin configuring your VPN access.

The VPN ensures secure communication between your local machine and the cloud resources, protecting your data and maintaining network security throughout your development work.





Step 9: Create VPN Peer Configuration

1

Add Peer

Click "Add Peer" and leave all default settings on the first page

2

Configure Public Key

Accept the automatically generated public key configuration

3

Set Network Options

Choose NAT option for Allowed IPs and proceed

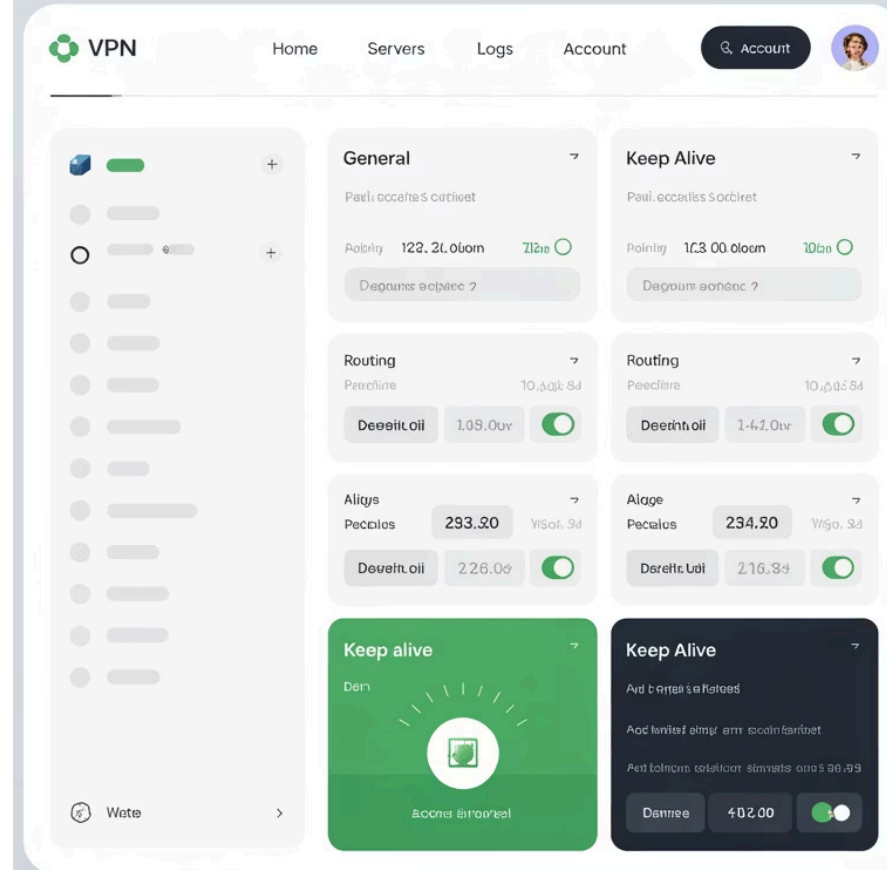
Step 10: Complete VPN Setup

Endpoint Configuration

On the "Endpoint & Keep Alive" page, accept the default values and click next. These settings ensure reliable VPN connectivity.

Peer Naming

Give your peer a descriptive name (like "yourname-hackathon") and click "Add" to create the configuration.



Current Configuration

Download the latest settings for seamless integration.

Home >

Settings >

Logs >

Help >

Sign Out >

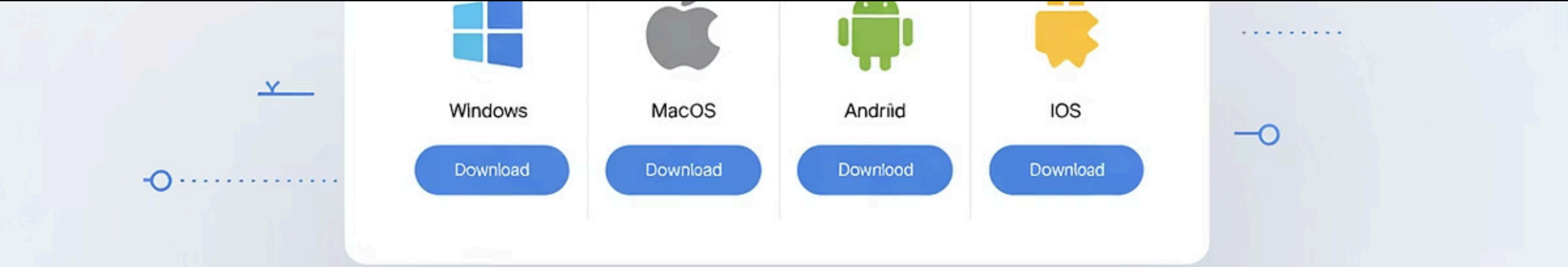


Download Configuration File

Step 11: Download VPN Configuration

 **Critical Step:** Click "Download Config" to save the configuration file as "yourname-hackathon.conf" on your computer. You'll need this file to connect to the VPN.

Store the configuration file in a safe location as it contains your unique VPN credentials. This file will be imported into your WireGuard client to establish the secure connection.



Step 12: Install WireGuard Client

If you haven't already, install WireGuard on your computer from <https://www.wireguard.com/install/>. WireGuard is available for Windows, macOS, Linux, and mobile platforms.

Choose the appropriate version for your operating system and follow the installation instructions. The lightweight VPN client will manage your secure connection to the hackathon infrastructure.

Step 13: Return to Create Virtual Machine

Navigate back to the "hackathon 2025" cloudspace. In the top menu, click **"Create VM"** to begin setting up your development virtual machine.

This virtual machine will serve as your primary development environment, providing you with a Ubuntu server where you can install tools, run applications, and develop your hackathon project.

console.sti.go.ug/customer/sti_industry___/cloudspaces/dWcia2hLWRpADeMDAaQYOMA/vms

console.sti.go.ug STI Industry 4.0 TEST USER ONE

DISABLE CLOUDSPACE DELETE CLOUDSPACE CREATE VM

Cloudspaces → hackathon 2025 → Virtual machines

Details

Name	hackathon 2025	Default External IP	41.220.3.53	Type	public
Status	DEPLOYED	Subnet	10.100.100.0/24	Default gateway	10.100.100.1
Location	ug-klc-001-001	Local domain	hackathon2025		

+ new note

VMS INGRESS DNS WIREGUARD VPNS EXTERNAL NETWORKS MORE

Search

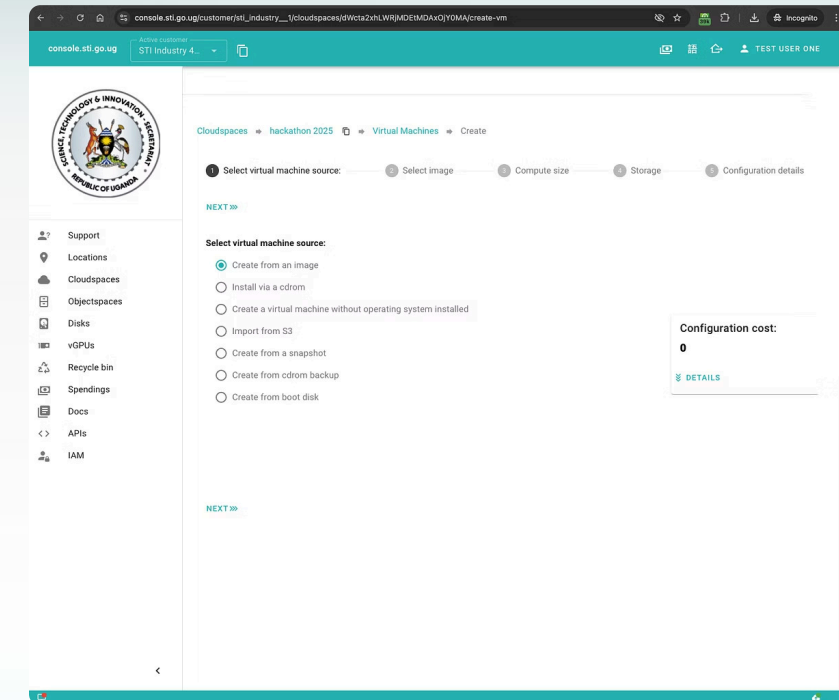
☐	Virtual machine	vCPUs	Memory (MB)	Storage (GB)	IP address	Status	Latest backup	Actions
☐	pass-ms	2	4096	150	10.100.100.2	RUNNING		II
☐	pass-wk01	4	8192	1124	10.100.100.3	RUNNING		II
☐	dohplay-01	2	4096	200	10.100.100.4	RUNNING		II
☐	testvm	2	2048	60	10.100.100.5	RUNNING		II

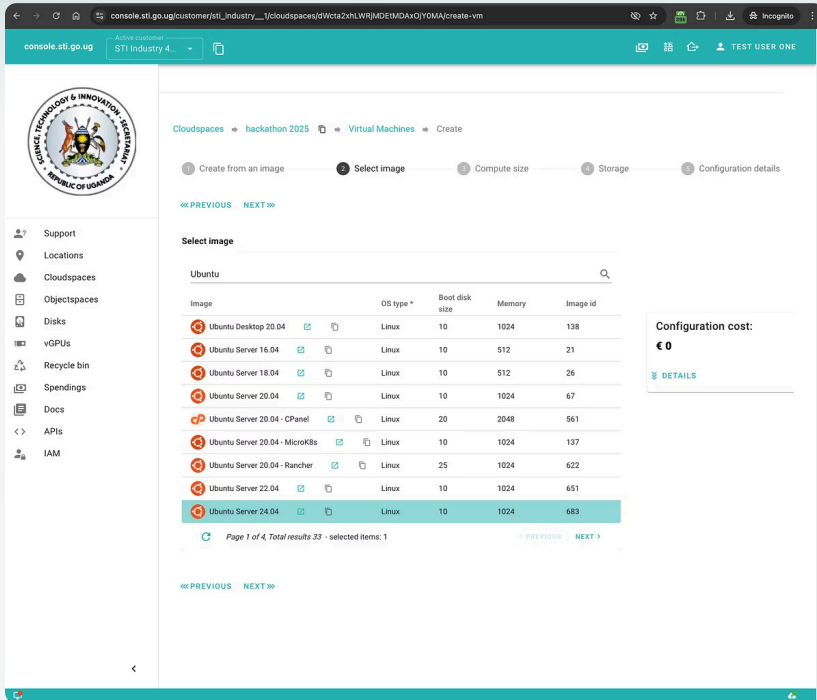
Page 1 of 1, Total results 4

Step 14: Choose VM Creation Method

On the VM creation page, select **"Create from an image"** and click next. This option allows you to start with a pre-configured operating system image, saving setup time.

Creating from an image provides a clean, standardised environment that's optimised for development work and compatible with the hackathon infrastructure.





Step 15: Select Operating System

Choose "**Ubuntu 24.04**" from the available images and click next. Ubuntu 24.04 LTS provides a stable, well-supported Linux environment perfect for development work.

This image comes pre-configured with essential development tools and libraries, providing an excellent foundation for various programming languages and frameworks commonly used in hackathons.

Step 16: Configure VM Resources

2

vCPU Cores

Processing power for compilation and development tasks

2048

RAM (MB)

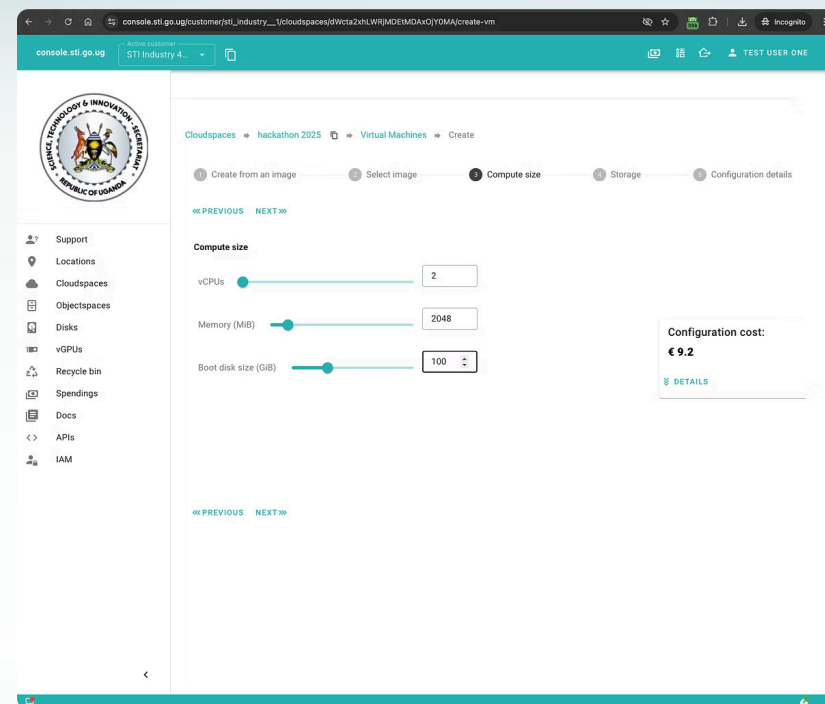
Memory allocation for running applications and services

1000

Disk (GB)

Storage space for code, dependencies, and project files

These resource allocations provide sufficient computing power for most hackathon projects whilst ensuring optimal performance across the shared infrastructure.



Step 17: Configure Storage and Networking

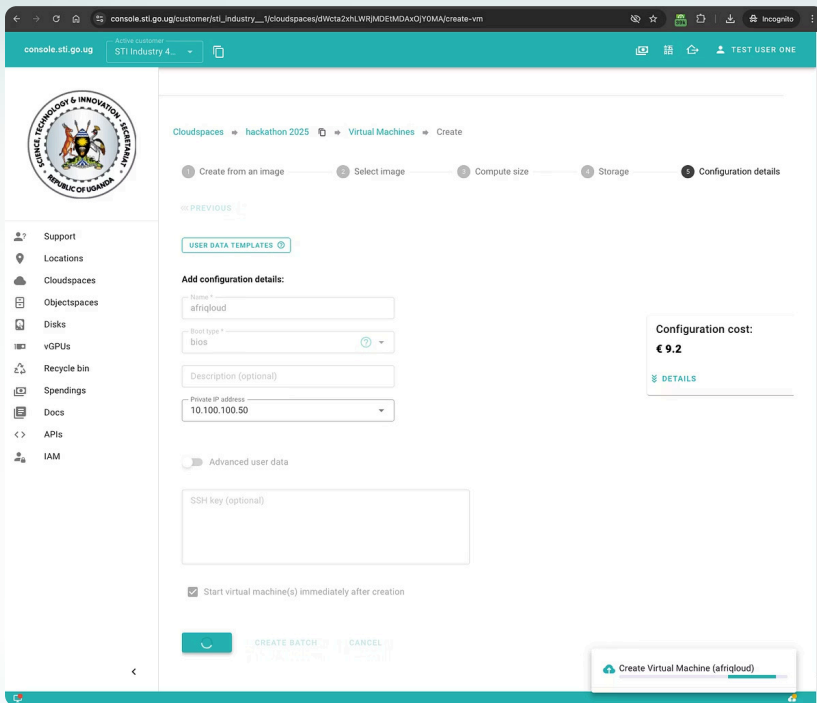
Data Disk Settings

Leave the data disk configuration as default and click next. The default settings provide adequate additional storage for your project needs.

Network Configuration

You'll configure the IP address and networking settings on the next page to ensure proper connectivity within the hackathon environment.





Step 18: Name and IP Configuration

VM Name

Give your VM a descriptive name like "teamname-dev-vm" for easy identification

IP Address Range

Select an IP address between
10.100.100.50 - 10.100.100.200

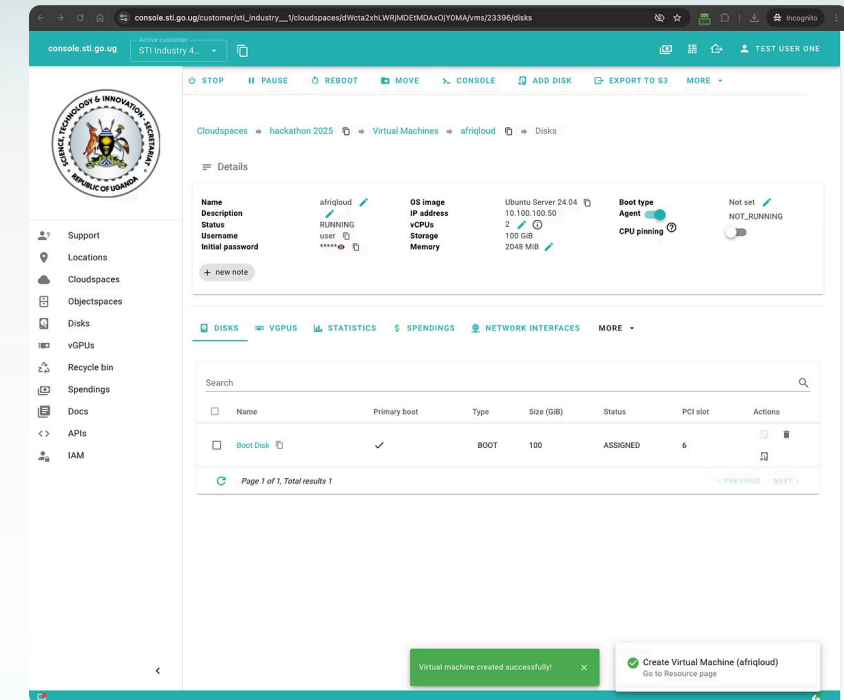
Default Settings

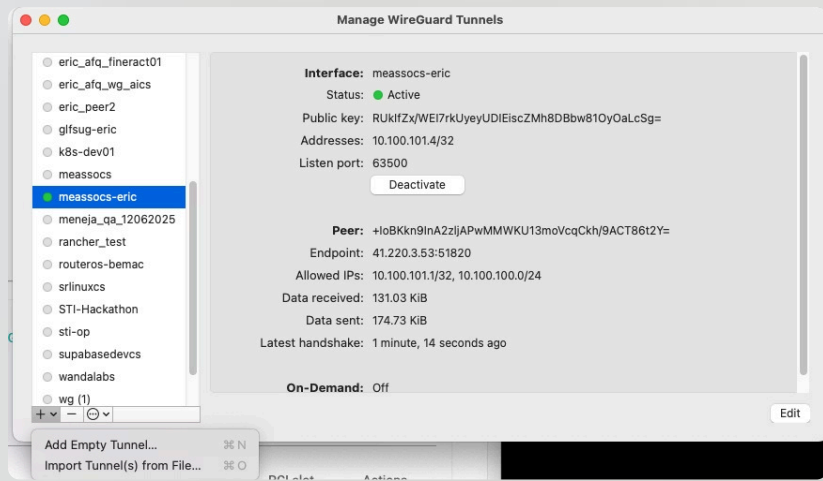
Leave all other settings as default and click "Create"

Step 19: VM Creation Process

The virtual machine creation process will take several minutes to complete. During this time, the system is provisioning your Ubuntu server, configuring networking, and preparing the development environment.

You can monitor the progress in the console. Once the status changes to "Running", your VM is ready for use and you can proceed to establish an SSH connection.





Step 20: Prepare for SSH Connection

Before connecting to your VM, you'll need to activate your WireGuard VPN using the configuration file you downloaded earlier. Import the .conf file into your WireGuard client and activate the connection.

The default login credentials are visible on the VM's console page: username is `user` and the password is your VM's unique ID displayed on the console.

Step 21: Connect via SSH

Connect to your VM using SSH. For Linux/Mac, use the terminal; for Windows, use PowerShell, Windows Terminal, or PuTTY.

```
ssh user@10.100.100.50
```

Replace the IP address with your VM's assigned IP. When prompted, enter the initial password (VM ID) shown on the console page. For security, immediately change your password using the `passwd` command after logging in.



A blurred background image of a computer monitor displaying a terminal window. The terminal shows a command prompt followed by the text "Changing password to" and then a password prompt with the text "newpassword123" and a green cursor.

```
Changing password to  
"newpassword123"
```

Step 22: Secure Your VM

1

Change Password

Run `passwd` to set a strong, unique password

2

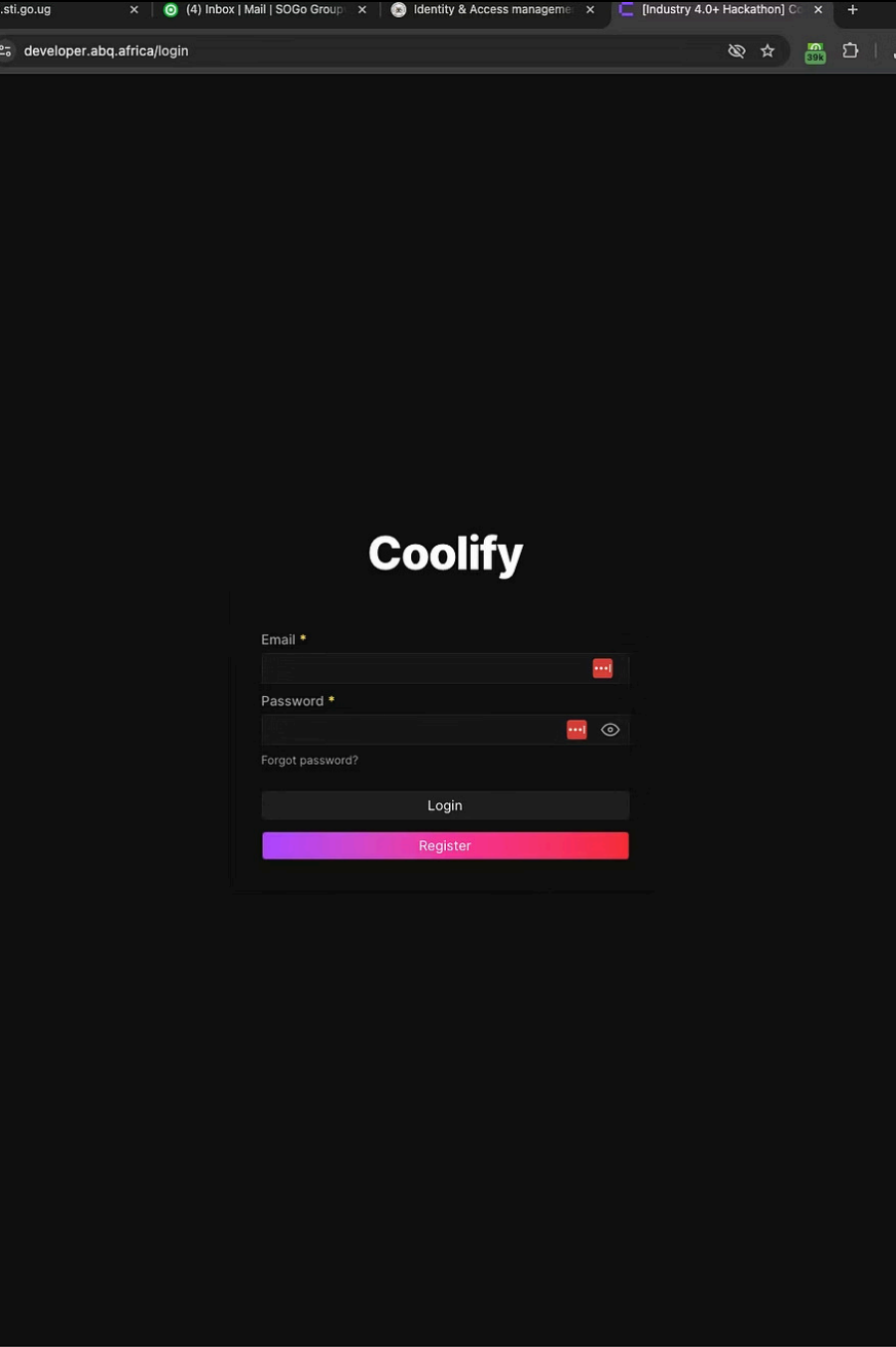
Gain Root Access

Use `sudo -i` when administrative privileges are needed

3

Install Software

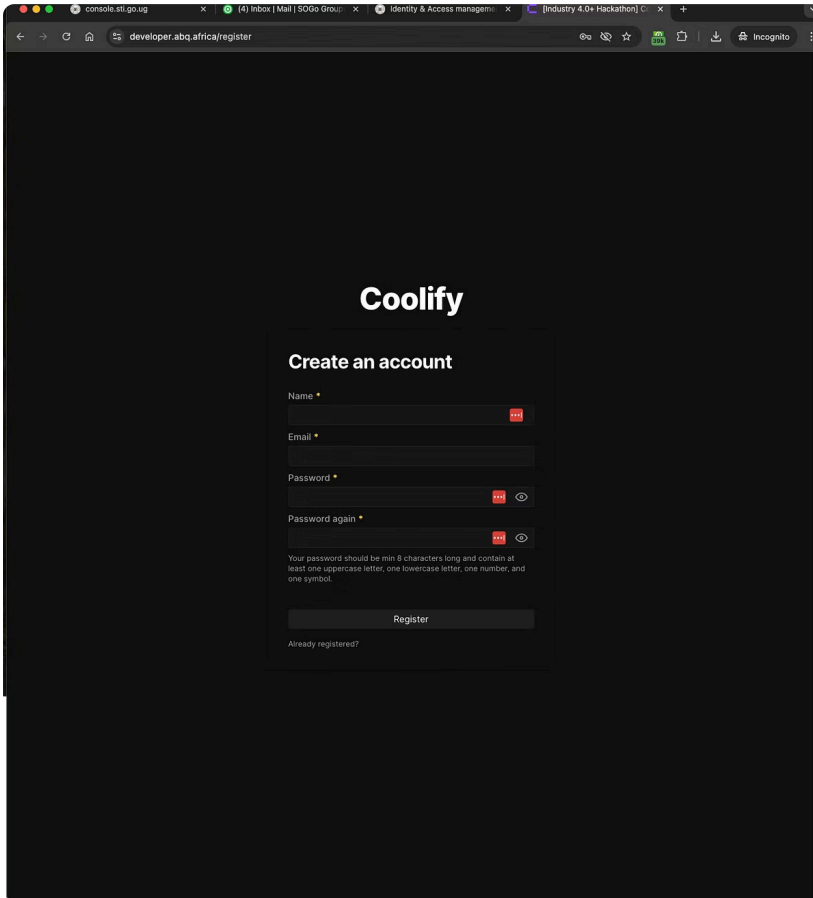
Install development tools and libraries required for your project



Step 23: Introduction to Coolify

Your VM will be connected to Coolify, a powerful self-hosted deployment platform that simplifies application deployment. Coolify allows you to deploy applications directly from Git repositories with minimal configuration.

Access Coolify at <http://developer.abq.africa> to streamline your application deployment process throughout the hackathon.



Step 24: Register on Coolify Platform

Navigate to <https://developer.abq.africa> and select "Register" to create your Coolify account. Fill in your details and password, then click register.

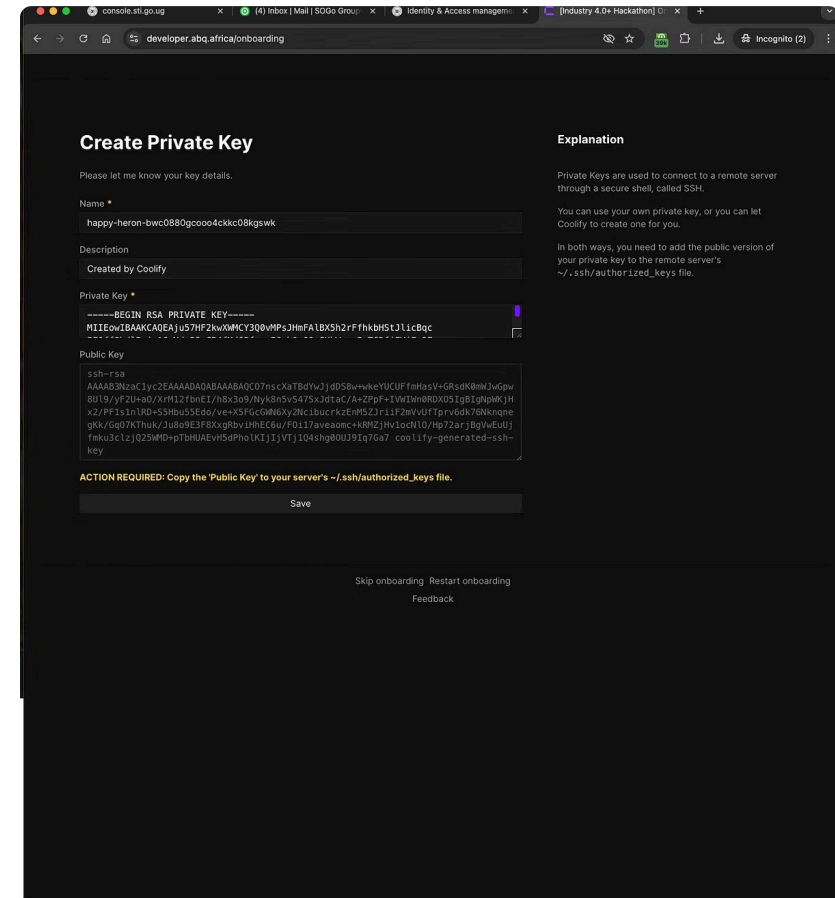
After registration, you'll be redirected to the landing page where you can click "Get Started" to begin configuring your deployment environment for easy application hosting.

Step 25: Configure Coolify Server Connection

Click "Next" under "Self-hosting with superpowers" and select "Remote server" to add your VM to Coolify's deployment system.

On the SSH Key page, select "No (create one for me)" to generate a new SSH key pair for secure server communication.

📄 Copy the generated public key - you'll need to add this to your VM's authorised keys for secure access.

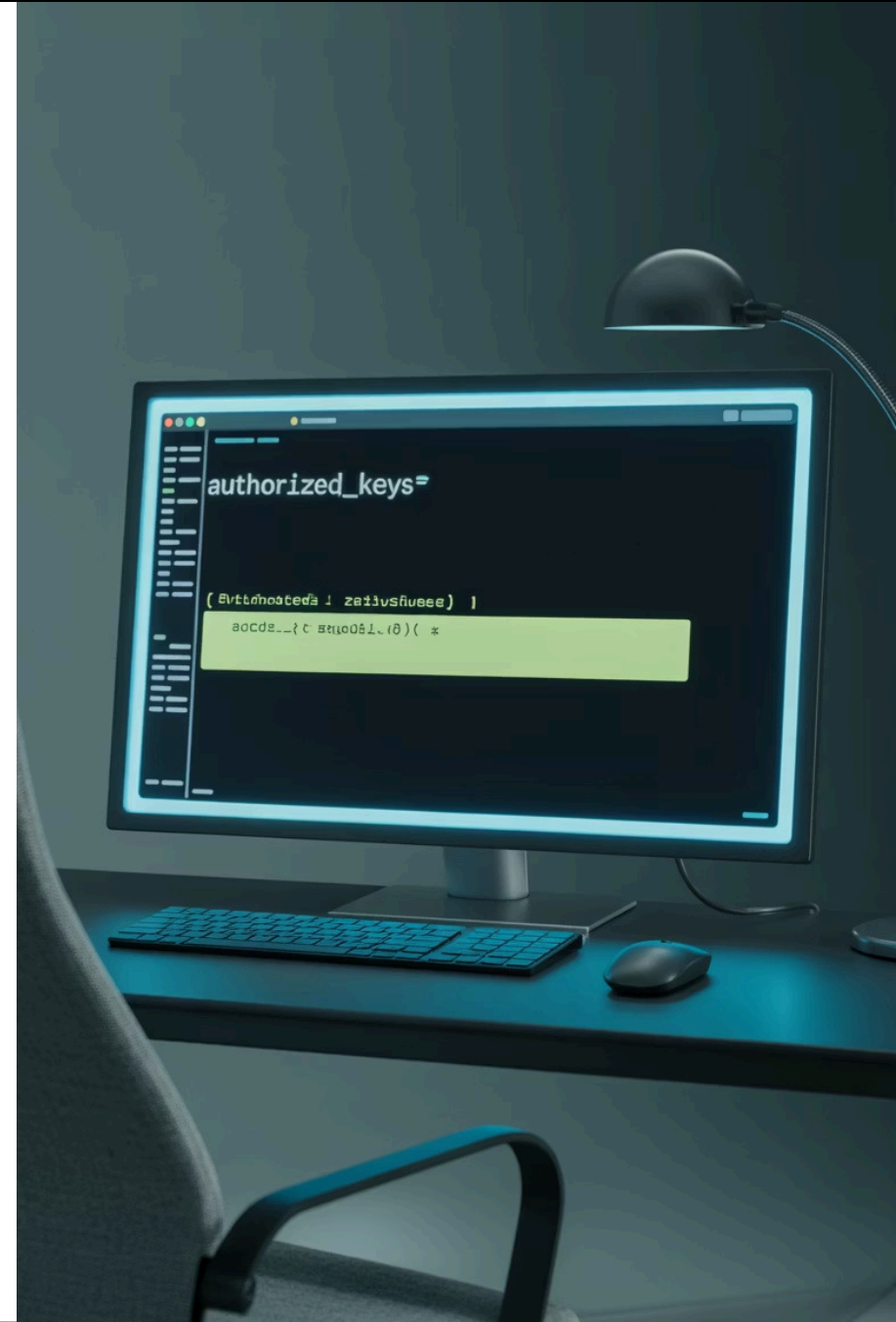


Step 26: Configure SSH Access

Connect to your VM via SSH and switch to root user with `sudo -i`. Open the authorised keys file using:

```
nano /root/.ssh/authorized_keys
```

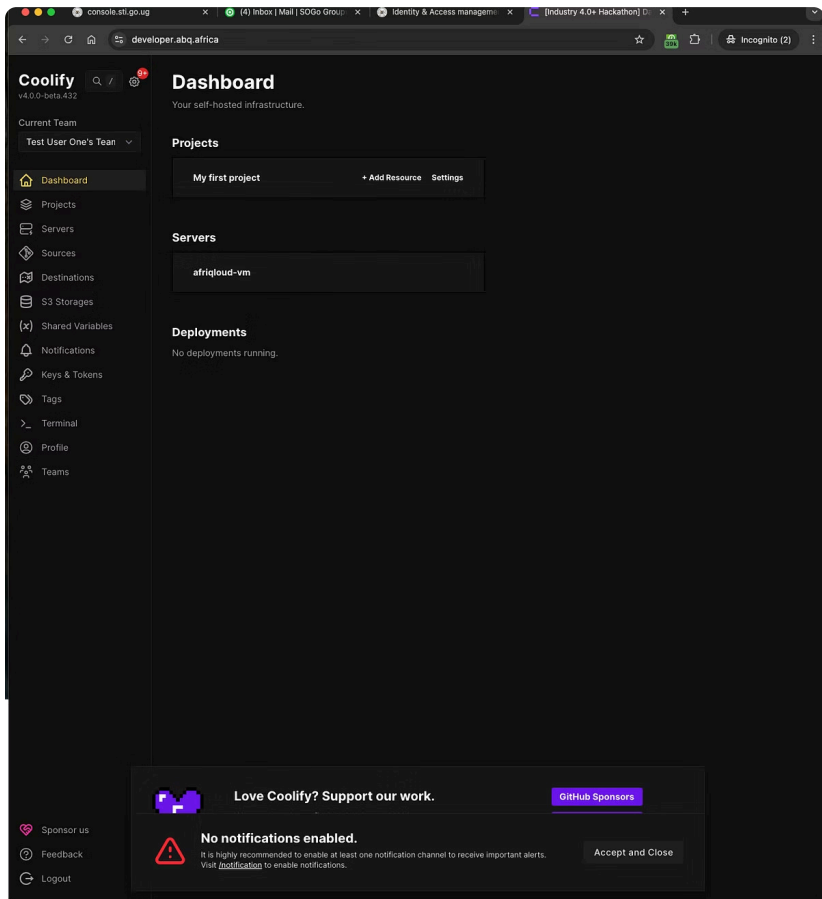
Paste the public key from Coolify at the end of the file and save. This enables Coolify to securely connect to your VM for automated deployments.



Return to Coolify and click "Save". Name your server (like "teamname-vm"), set your VM's IP address from the console page, and click "Continue".

The image shows a web browser window displaying the Docker Desktop onboarding page. The browser's address bar shows the URL 'developer.abq.afriqai.com/boarding'. The page has a dark theme. At the top, there are several browser tabs open. The main content area is divided into two columns. The left column is titled 'Validate & Configure Server' and contains a paragraph asking if the user is okay with the setup, followed by a large blue button labeled 'Let's do it'. The right column is titled 'Explanation' and contains text about installing the latest Docker Engine, the minimum version being 24, and a link to the documentation. At the bottom of the page, there are links for 'Skip onboarding', 'Restart onboarding', and 'Feedback'.

[illegible]



You're Ready to Build!

VM Ready

Your Ubuntu development environment is configured and accessible via SSH

VPN Connected

Secure WireGuard connection established for safe resource access

Coolify Integrated

Deployment platform ready for easy application hosting and management

Your development environment is fully configured! Create a new project in Coolify by clicking "Create new project!" then "Let's do it!" Your server will be listed under "Servers" where you can manage deployments throughout the hackathon. Happy coding!